Mac Forensics

One Apple, many operating systems

- **macOS** – Mac desktops and laptops (MacBook Air/Pro, iMac, Mac mini, Mac Studio, Mac Pro)
- **iOS** – iPhone only. iPod touch ended at iOS 15 and was discontinued in 2022
- **iPadOS** – iPad, iPad mini, iPad Air, iPad Pro. Split from iOS in 2019 (iPadOS 13)
- **watchOS** – Apple Watch. **tvOS** – Apple TV HD (2015) and later; older Apple TVs ran “Apple TV Software”
- **visionOS** – Vision Pro. **bridgeOS** – runs on the T2 chip inside 2018–2020 Intel Macs

- Same company, same kernel (Darwin/XNU), same artifact families (plists, SQLite, Biome, Unified Logs). Different version numbers, different support lists, different acquisition rules.

Current versions

Device	OS	Current public version	Oldest supported device
Mac	macOS 27 Golden Gate	27.0.1 (28 Sep 2026). Tahoe 26.7.1 still updated for Intel Macs	M1 Macs (2020) and MacBook Neo (A18 Pro). No Intel
iPhone	iOS 27	27.0.1. iOS 27.1 ships with iPhone Duo on 23 Oct. 27.2 in RC	iPhone 11 (2019), iPhone SE 2nd gen
iPad	iPadOS 27	27.0.x	iPad 9th gen, iPad mini 6, iPad Air 4. 2018–2020 Pros dropped
Apple Watch	watchOS 27	27.0.x	Series 9, Ultra 2, SE 3rd gen. Series 6–8 dropped
Apple TV	tvOS 27	27.0.x	Apple TV 4K 2nd gen (2021)
Vision Pro	visionOS 27	27.0.x	Vision Pro (M2) and Vision Pro (M5)

Recent macOS versions: forensic impact

macOS	Version	Forensic impact
Catalina	10.15 (2019)	Read-only System volume and separate Data volume. Artifact paths resolve through firmlinks.
Big Sur to Sonoma	11 to 14 (2020–2023)	Signed System Volume, APFS snapshots, Biome replaces most of KnowledgeC, heavier TCC privacy controls.
Sequoia	15 (2024)	iPhone Mirroring artifacts, Passwords app, more Apple Intelligence traces on M-series Macs.
Tahoe	26 (2025)	Year-based numbering begins. **Last macOS that runs on Intel Macs.** Liquid Glass UI, Apple Intelligence everywhere.
Golden Gate	27 (Sept 2026)	**Apple silicon only.** Last version with full Rosetta 2. AFP, Boot Camp and AirPort Utility removed. Intel Macs stay on Tahoe with security updates only.

OS changes that matter to investigators

- Do not assume one path, one database schema, or one timestamp pattern works across all recent macOS releases.
- The same artifact family can move, split, or lose fields after major OS upgrades.
- Always document the exact build number, hardware model, chip family, FileVault state, and user account context.
- When a commercial tool shows a result, verify at least one item by locating the source file and checking the raw record.

Hardware generation and acquisition consequences

Generation	Storage / security pattern	Collection consequence
Older Intel	Removable SATA/NVMe storage on many models	Dead-box removal may work. Still confirm FileVault and APFS.
Intel with T2	Controller and Secure Enclave tied to internal storage	Prefer live/booted acquisition when credentials are available. Dead storage access is limited.
Apple silicon	Integrated storage, sealed boot chain, Secure Enclave	User credentials and trusted boot state are often decisive. Plan live capture carefully.
Fusion Drive	SSD + HDD combined by CoreStorage/APFS logic	Acquire the logical set, not just the visible HDD. Missing tier can break reconstruction.

Mac iterations

Era	Years	What changed	What it means for an examiner
Motorola 68k	1984–1994	Original Macintosh. HFS file system.	Museum pieces. HFS/HFS+ epochs still show up in old media.
PowerPC	1994–2006	Mac OS 8/9, then Mac OS X (2001). HFS+.	Rare. Removable IDE/SATA drives, no encryption by default.
Intel	2006–2020	Boot Camp, EFI, FileVault 2 (2011), APFS (2017).	Removable SATA or NVMe on most models. Classic dead-box imaging works.
Intel + T2	2018–2020	T2 security chip: Secure Enclave, SSD controller, encrypted storage always on.	Storage removal gives ciphertext. Need credentials and a boot-based or live acquisition.
Apple silicon	2020–today	M1 to M5 (and A18 Pro in MacBook Neo). SoC with unified memory, SSD controller and Secure Enclave on one die.	No Target Disk Mode (use Share Disk), no Boot Camp, sealed boot chain. macOS 27 is Apple silicon only.

Intel vs Apple silicon

- **Where the keys live.** Intel: keys on the drive, protected by the FileVault password. Apple silicon: keys in the Secure Enclave, never leave the chip. Internal SSD is encrypted even with FileVault off.
- **How you boot it.** Intel: Cmd+R, Option-boot from USB, Target Disk Mode (T). Apple silicon: hold power for Options, Share Disk over SMB, DFU restore via Apple Configurator.
- **Who may change security.** Apple silicon ties SIP, external boot and kernel extensions to a volume owner (Secure Token user). No owner password, no changes.
- **Memory.** Unified memory shared with the GPU. A live RAM capture is the only way to get it.

The SSD

- Soldered NAND started with the 12" MacBook (2015) and the Touch Bar MacBook Pro (2016). Every Apple silicon MacBook, iMac, Mac mini and MacBook Neo has NAND packages soldered to the logic board.
- Only Mac Studio and Mac Pro use removable NAND modules, and even those are paired to that specific machine's Secure Enclave.
- The SSD **controller** is inside the T2 or M-series SoC. Every block is written encrypted with a key that exists only in that chip. FileVault merely wraps that key with the user password.
- Consequence: chip-off, board swap or a NAND reader produce ciphertext. There is no "pull the drive and image it" for a 2018+ Mac.
- What works instead: (1) live acquisition on the running, unlocked Mac; (2) boot the Mac into Recovery and use Share Disk or a collector such as Digital Collector; (3) obtain the password or recovery key first.
- Fusion Drive and Apple RAID iMacs (2012–2020) are two devices that must be acquired together.

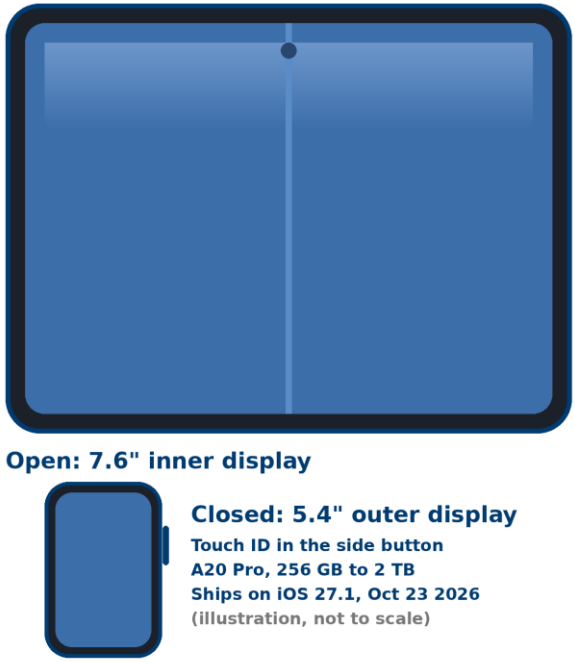
Why Apple devices are hard

- **Encryption by default.** FileVault on Macs, Data Protection on iPhone, keys sealed in hardware. The password is the evidence key.
- **The OS fights your tools.** SIP, Signed System Volume block access to user data unless the tool has Full Disk Access; Gatekeeper and notarization refuse unsigned binaries.
- **You need a Mac to examine a Mac.** APFS, FileVault, Apple silicon boot and keychain all parse best on Apple hardware. Few vendors boot Apple silicon at all.
- **Annual churn.** Every September artifacts move, rename or vanish (KnowledgeC to Biome, airport.preferences to known-networks, .sfl2 to .sfl3). Tools lag by months.
- **Privacy features.** Unified Log redaction, Lockdown Mode, Advanced Data Protection for iCloud, inactivity reboot on iPhone (back to BFU after 72 h), USB Restricted Mode.
- **Cloud is the other half.** iCloud Drive, Messages in iCloud and Photos may not be on the device at all; placeholders hold metadata only.

iPhone Duo (Sept 2026): Apple's first foldable

Forensic challenges

- Two screens, two UI states: artifacts, screenshots and app layouts differ folded vs unfolded. Document the state at seizure.
- Touch ID in the side button plus Apple Watch unlock; no Face ID. Plan the biometric window before the device locks.
- Ships on iOS 27.1 on 23 Oct 2026: expect weeks to months before extraction tools support it.
- Up to 2 TB of storage: full file system extractions become multi-hour, multi-hundred-GB jobs.
- Hidden under-display FaceTime camera and new Duo-only apps and streams (Smart Take, Duo Preview) mean new Biome and KnowledgeC stream names.



Every new iPhone, same questions

- **BFU or AFU?** Before First Unlock the data keys are not in memory; after unlock they are. A Duo that was folded shut and never powered off is still AFU. Keep it powered.
- **Inactivity reboot.** Since iOS 18.1 an iPhone reboots itself after about 72 hours without unlock and drops back to BFU. The clock starts at seizure.
- **USB Restricted Mode and Lockdown Mode.** Data over Lightning/USB-C is cut off one hour after lock unless a trusted computer connects. A Faraday bag does not stop this timer.
- **Stolen Device Protection.** Away from trusted locations, changing Apple ID or passcode needs Touch ID plus a one-hour delay. Affects consent-based collections too.
- **Tool reality:** Cellebrite, GrayKey and Magnet are among the only tools that support Mac.


Apple macOS History
BIG CATS
Cheetah
Puma
Jaguar
Panther
Tiger
Leopard
Snow Leopard
Lion
Mountain Lion
deepakkeswani.com
CALIFORNIA LOCATIONS
Mavericks
Yosemite
El Capitan
Sierra
High Sierra
Mojave
Catalina
deepakkeswani.com
Big Sur
Monterey
Ventura
Sonoma
INTEL & ARM
CPU SUPPORT

Timestamps on Mac

- Unix epoch – Number of seconds from 1/1/1970 midnight.
- Mac Absolute Time – Number of seconds from 1/1/2001 midnight.
- HFS+ Time – Number of seconds from 1/1/1904 midnight.

https://github.com/pstirparo/utils/blob/master/epochalypse.py

Timestamps on Mac

https://www.digital-detective.net/dcode/

https://gchq.github.io/CyberChef/

Mac timestamp quick reference

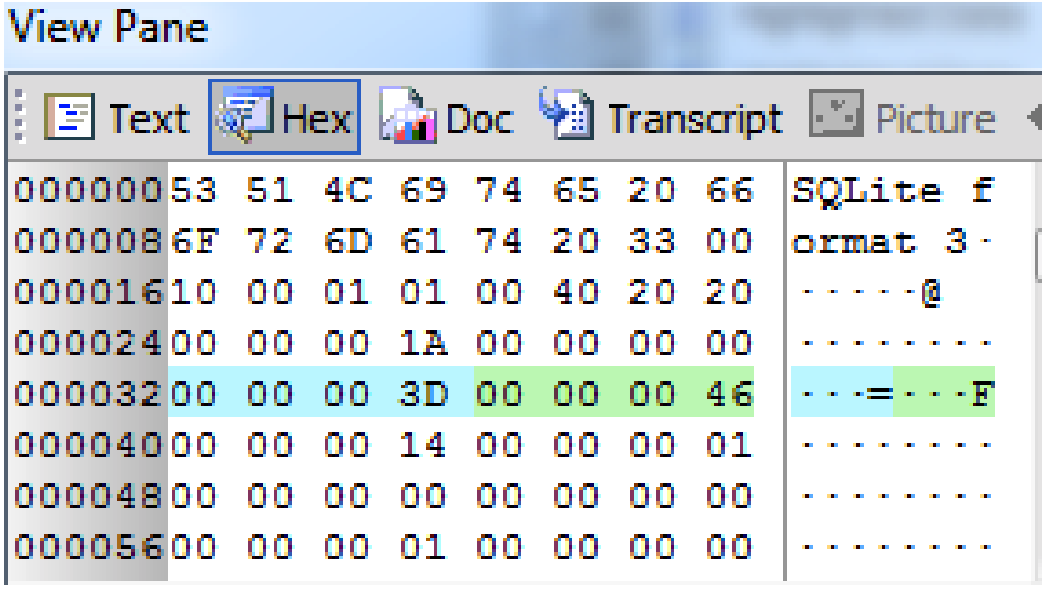
Timestamp family	Epoch	Common places
Unix	1970-01-01 UTC	APFS metadata in many tools, log exports, browser artifacts, shell history depending on source.
Mac Absolute / Cocoa	2001-01-01 UTC	KnowledgeC, Biome naming/records, Core Data databases, many Apple application timestamps.
HFS+	1904-01-01 UTC	Legacy HFS+ metadata and older Mac artifacts.
WebKit / Chrome	1601-01-01 UTC, microseconds	Chromium history, cookies, login data and other browser databases.

Databases on Mac systems

Most of the software on Mac uses SQLite databases to store various type of data – user generated data, settings, logs.

SQLite databases are complex systems that can store multiple layers of variables within.

There are different types of SQLite databases. Mac uses the SQLite format 3 type:


View Pane
Text
Hex
Doc
Transcript
Picture
00000053 51 4C 69 74 65 20 66 SQLite f
00000860 72 6D 61 74 20 33 00 ormat 3·
00001610 00 01 01 00 40 20 20 ·····@
00002400 00 00 1A 00 00 00 00 ·····
00003200 00 00 3D 00 00 00 46 ···=···F
00004000 00 00 14 00 00 00 01 ·····
00004800 00 00 00 00 00 00 00 ·····
00005600 00 00 01 00 00 00 00 ·····

The constant signature means that the SQLite database can be carved.

The file can be viewed by various tools, such as DB Browser for SQLite - https://sqlitebrowser.org/

Sometimes, other files can be part of the SQLite database - .shm – Shared Memory File and .wal – Write Ahead Log.

SQLite file/database does not have a constant extension and can be used by multiple tools/OS/FS in various formats with various extensions. The best way to tell it's an SQL file is to view the header, or open it in a DB View tool.

SQLite WAL and SHM files

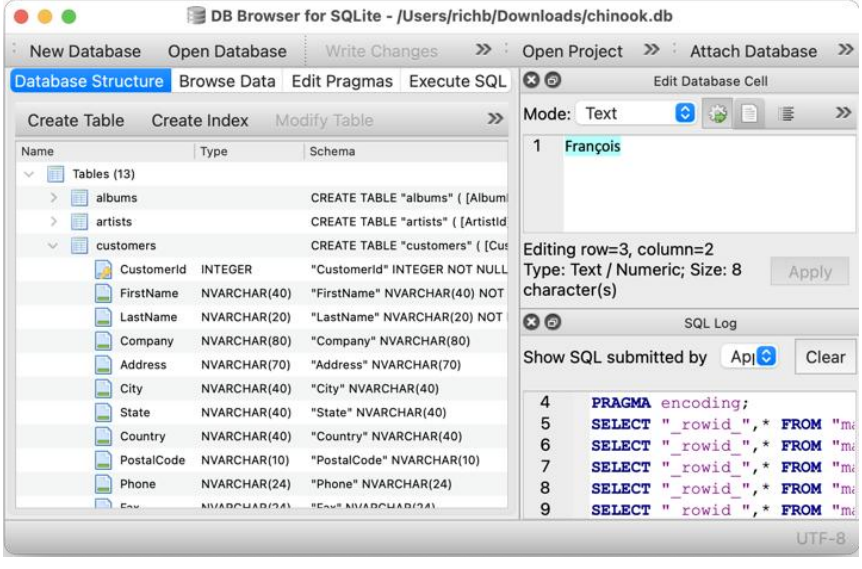
- The .wal file can contain records that are newer than the main database file.
- The .shm file is not normally an evidence target by itself, but it confirms shared-memory WAL activity.
- Opening a live database can checkpoint or alter WAL state. Work on forensic copies only.
- For application artifacts, the newest record may be in the WAL while the main database looks stale.

Common macOS SQLite databases

Artifact	Typical source path	Investigative value
Safari History	~/Library/Safari/History.db	Visited URLs, visit times, redirect chains, browser timeline.
Messages	~/Library/Messages/chat.db	iMessage/SMS chats, handles, attachments, chat joins.
KnowledgeC	~/Library/Application Support/Knowledge/knowledgeC.db	Application focus, device/user activity and pattern-of-life data.
TCC	~/Library/Application Support/com.apple.TCC/TCC.db	Privacy permissions granted or denied to apps.
Notes	~/Library/Group Containers/group.com.apple.notes/NoteStore.sqlite	Notes metadata and content, with schema and encryption caveats.

Tool example: DB Browser for SQLite

- Useful for first-pass schema review, SQL queries and manual validation.
- Never open the original evidence database directly.
- Document exact query used and the source files included in the copy set.

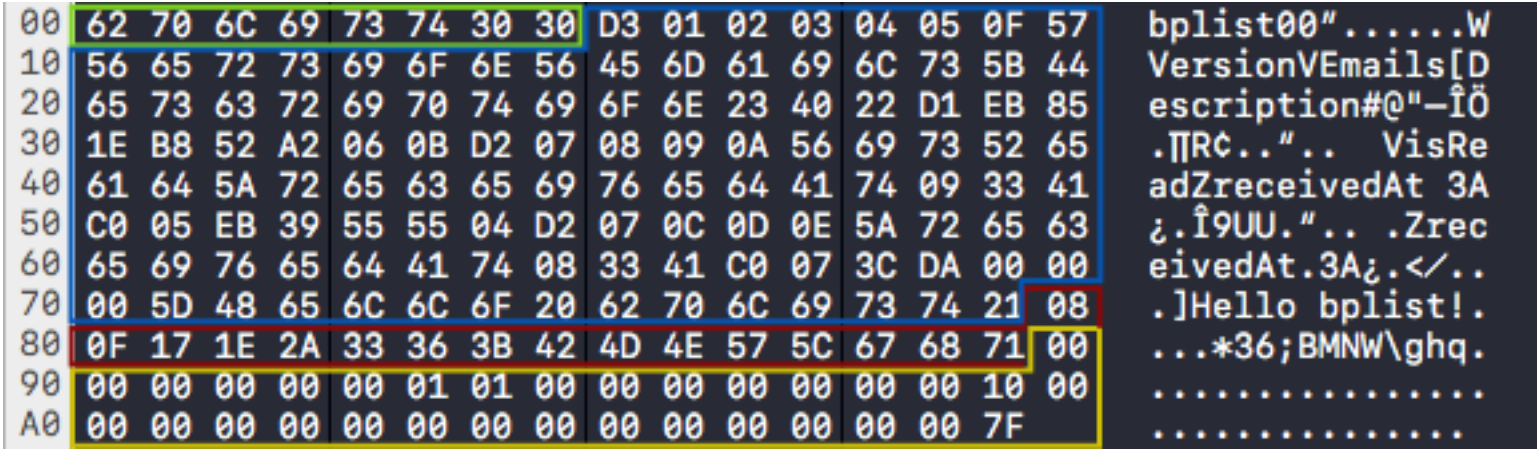


Plists

- A **property list** (.plist) is a small settings file.
- Inside it is a list of **keys and values**: “LastOpened = 2026-10-13 21:16”, “RecentFiles = [a, b, c]”, “ShowHiddenFiles = true”.
- Windows keeps this kind of data in the Registry, one big central database. Apple spreads it over thousands of small plist files, one per app or setting.
- Three types you will meet: **XML** (human readable, starts with <?xml), **binary** (starts with bplist00, the default since 2002), and **nested archives** (NSKeyedArchiver: a plist whose values are more plists, needs a parser).
- Why we care: plists hold recent files, Wi-Fi networks, mounted volumes, login items, Dock contents, app accounts, last-run times and much more.

P (property) Lists

Binary Plists – “bplist00”



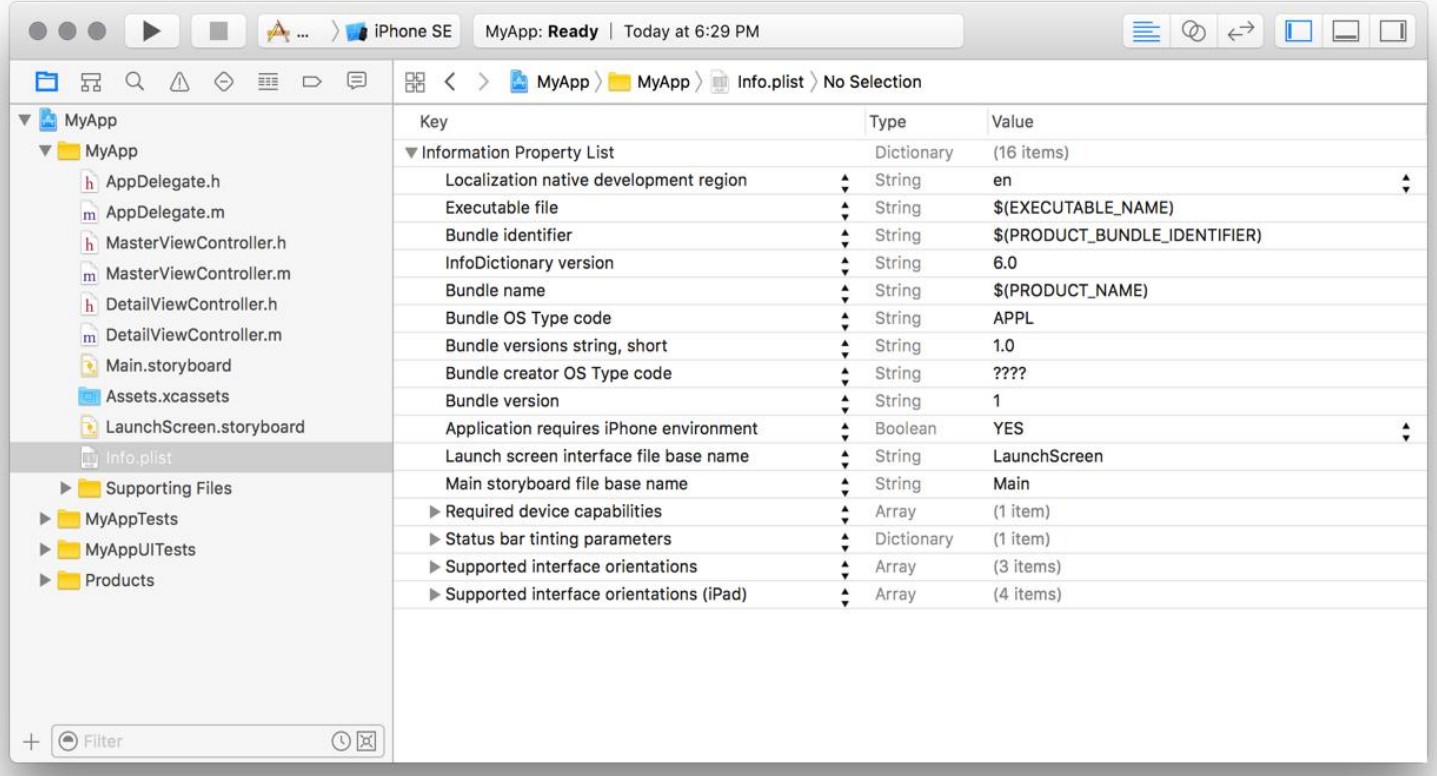
```
00 62 70 6C 69 73 74 30 30 D3 01 02 03 04 05 0F 57   bplist00"......W
10 56 65 72 73 69 6F 6E 56 45 6D 61 69 6C 73 5B 44   VersionVEmails[D
20 65 73 63 72 69 70 74 69 6F 6E 23 40 22 D1 EB 85   escription#@"—ÎÖ
30 1E B8 52 A2 06 0B D2 07 08 09 0A 56 69 73 52 65   .¶R¢..".. VisRe
40 61 64 5A 72 65 63 65 69 76 65 64 41 74 09 33 41   adZreceivedAt 3A
50 C0 05 EB 39 55 55 04 D2 07 0C 0D 0E 5A 72 65 63   ¿.Î9UU."..  .Zrec
60 65 69 76 65 64 41 74 08 33 41 C0 07 3C DA 00 00   eivedAt.3A¿.</..
70 00 5D 48 65 6C 6C 6F 20 62 70 6C 69 73 74 21 08   .]Hello bplist!.
80 0F 17 1E 2A 33 36 3B 42 4D 4E 57 5C 67 68 71 00   ...*36;BMNW\ghq.
90 00 00 00 00 00 01 01 00 00 00 00 00 00 00 10 00   ................
A0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 7F      ................
```

Reading plists off the Mac

- **Python plistlib** (built in, every OS): loads XML and binary plists into dictionaries. Two lines of code, no install.
- **plistutil** (libplist): command line converter for Windows and Linux.
- **Plist Editor Pro** (Windows GUI) if you want to click, but verify its date decoding.
- **Forensic suites**: Magnet AXIOM, X-Ways, Cellebrite Physical Analyzer and Inspector parse plists and NSKeyedArchiver blobs natively.

Plist locations worth checking

Artifact	Typical path	What it may show
User preferences	~/Library/Preferences/*.plist ~/Library/Containers/<bundle>/Data/Library/Preferences/	Application settings, recent values, identifiers and account state.
Global preferences	~/Library/Preferences/.GlobalPreferences.plist	Locale, UI, keyboard and behavior settings.
Dock items	~/Library/Preferences/com.apple.dock.plist	Pinned apps, recent apps, persistent file/network items.
Login / background items	Ventura+: /private/var/db/com.apple.backgroundtaskmanagement/BackgroundItems-v*.btm Monterey and earlier: ~/Library/Application Support/com.apple.backgroundtaskmanagementagent/backgrounditems.btm	User and managed background items (persistence).
Launch agents	~/Library/LaunchAgents/*.plist	User persistence entries and scheduled execution.


iPhone SE
MyApp: Ready | Today at 6:29 PM
MyApp 〉 MyApp 〉 Info.plist 〉 No Selection
MyApp
MyApp
AppDelegate.h
AppDelegate.m
MasterViewController.h
MasterViewController.m
DetailViewController.h
DetailViewController.m
Main.storyboard
Assets.xcassets
LaunchScreen.storyboard
Info.plist
Supporting Files
MyAppTests
MyAppUITests
Products
Filter
Key
Type
Value
Information Property List
Dictionary
(16 items)
Localization native development region
String
en
Executable file
String
$(EXECUTABLE_NAME)
Bundle identifier
String
$(PRODUCT_BUNDLE_IDENTIFIER)
InfoDictionary version
String
6.0
Bundle name
String
$(PRODUCT_NAME)
Bundle OS Type code
String
APPL
Bundle versions string, short
String
1.0
Bundle creator OS Type code
String
????
Bundle version
String
1
Application requires iPhone environment
Boolean
YES
Launch screen interface file base name
String
LaunchScreen
Main storyboard file base name
String
Main
Required device capabilities
Array
(1 item)
Status bar tinting parameters
Dictionary
(1 item)
Supported interface orientations
Array
(3 items)
Supported interface orientations (iPad)
Array
(4 items)

MacOS File System Domains

The Domains are used to separate files based on their usage. This helps maintain access controls, to make sure files are not intentionally altered.

User Domain

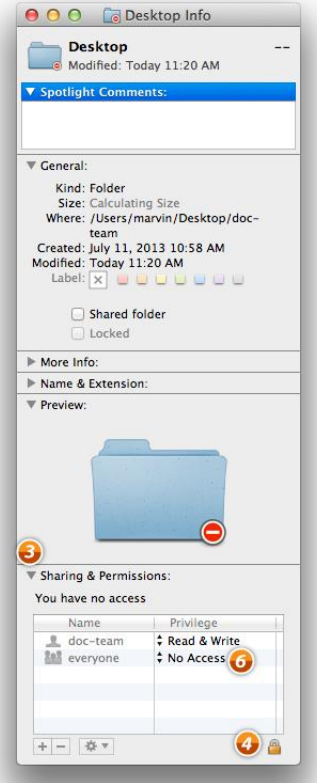
User-specific data, controlled by each user. Pictures, documents, music, etc.

MacOS File System Domains

User Domain

When a user is logged on, they can only access their home directory in /Users folder. By default, other user profiles are not accessible, except the “public” folder.

The “Library” folder in each user profile contains app-specific data (hidden by default)



MacOS File System Domains

Local Domain

Contains the Applications folder and the main Library folder.

Dedicated to files that are shared among the various users of the device.

MacOS File System Domains

System Domain

Stores Apple-specific system software. It contains the System Library folder – this is where we can find many of the Plist files.

MacOS File System Domains

Network Domain

Local network resources and apps – usually controlled by net admin.

Systems like NAS drives, Time capsules, network printers.

APFS structure to document

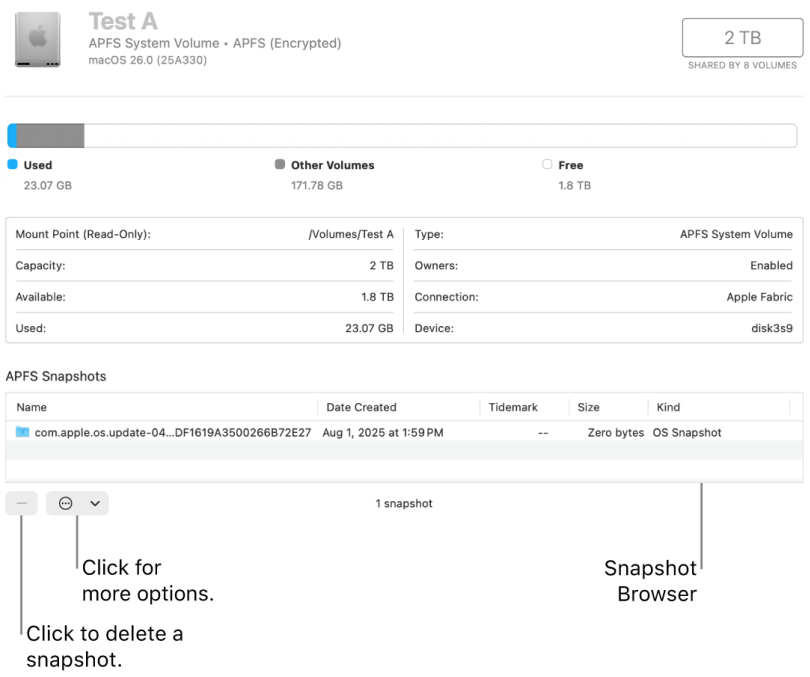
Layer	What it means	Why it matters
Physical disk	Real internal or external device	May not expose user data directly on T2/Apple silicon.
Container	APFS space-sharing pool	Can hold multiple volumes and snapshots. Image the right level.
Volume group	System + Data pairing	Recent macOS stores OS and user data separately.
Snapshots	Read-only historical states	May preserve files or metadata that are gone from the live view.

Catalina and later: System/Data split

- macOS 10.15 introduced a dedicated read-only System volume and a separate Data volume for mutable files.
- The user sees a merged filesystem, but the acquisition may expose separate volumes.
- Firmlinks/HardLinks make directories appear in familiar locations while data actually resides on the Data volume.
- Forensic tools must understand the split or the examiner may miss user files, system files or historical snapshots.

APFS snapshots in Disk Utility

- Snapshots are read-only copies of the parent APFS volume at a point in time.
- They can preserve previous states of files, metadata and directories.
- Confirm snapshot name, date created, UUID/XID and whether the tool processed it.



User home high-value targets

- ~/Desktop, ~/Documents, ~/Downloads and ~/Pictures - user-created and downloaded files.
- ~/Library - application data, caches, preferences, saved states, logs and databases.
- Hidden files such as .zsh_history, .bash_history, .ssh, .gitconfig and tool-specific configs.
- Group Containers and Containers - sandboxed app data, especially Notes, Safari, Mail, iCloud and third-party apps.

Mac Acquisition

- **Intel Macs up to about 2017** had a standalone SATA drive or NVMe blade. For a dead-box case, removing the drive is still the cleanest option (see www.ifixit.com for teardowns).
- **Soldered NAND** started with the 12" MacBook (2015) and the Touch Bar MacBook Pro (2016). **T2 Macs (2018–2020)** and **every Apple silicon Mac (2020+)** add hardware-bound encryption on top.
- On those machines removing storage is pointless, not just difficult: the NAND holds ciphertext and the key never leaves the chip.
- Practical paths today: live acquisition on the unlocked Mac, or boot the Mac itself (Recovery, Share Disk, or a boot collector such as Cellebrite Digital Collector) with the user password or recovery key.

Mac Acquisition: encryption

- **Legacy FileVault** (10.3 to 10.6): encrypted only the user's home folder as a sparse bundle.
- **FileVault 2** (10.7 Lion onward): full volume encryption, XTS-AES-128 with a 256-bit key. On T2 and Apple silicon the hardware already encrypts everything; FileVault adds the password wrap.
- Ways in, in order of likelihood: the user password; the **personal recovery key** (24 characters, shown at setup); an **institutional recovery key** or **MDM-escrowed key** on managed Macs; a key escrowed to **iCloud** by the user.
- No key, no data. There is no known bypass of a healthy FileVault 2 volume. A powered-on, unlocked Mac is therefore worth more than any imaging kit.

Mac Acquisition

SIP – System Integrity Protection





Mac Acquisition: SIP

- **System Integrity Protection** (since 10.11 El Capitan) stops even root from modifying protected system locations, loading unsigned kernel extensions or attaching a debugger to Apple processes.
- Since Big Sur the whole System volume is also a **Signed System Volume** (SSV): a cryptographic seal that the Mac verifies at boot.
- Why it matters to us: live collection tools that need kernel access (memory capture, raw disk reads) may be blocked while SIP is on, and some agent-based collectors require Full Disk Access in TCC instead.
- SIP state can only be changed from Recovery. Changing it is a modification of the evidence system: document it, and prefer a boot-based or agent-based collection that does not require it.

How to check or change SIP

Check (any Mac, no reboot)

- Terminal: **csrutil status**

Intel Mac

- Restart, hold **Cmd + R** until the Apple logo. Utilities > Terminal, then **csrutil disable** (or enable). Restart.

Apple silicon Mac

- Shut down, then **hold the power button** until “Loading startup options”. Choose Options > Continue, pick a volume and enter an **admin (volume owner) password**. Utilities > Terminal, then **csrutil disable**. Restart.
- On Apple silicon this also drops the security policy to Reduced Security. Note it in your report and restore it when done.

Mac Acquisition: Fusion Drive (2012–2020)

- Two physical devices presented as one: a 1, 2 or 3 TB hard drive plus a small SSD (128 GB at launch; the 1 TB model was later cut to 24 GB, then 32 GB).
- Not RAID 0. It is **tiered storage**: macOS keeps frequently used blocks on the SSD and moves cold blocks to the HDD, so any given file may live on either device.
- Glued together by **Core Storage** (2012–2018) or **APFS Fusion** (Mojave 10.14 onward).
- Forensic consequence: imaging only the HDD gives you a broken file system with holes where the hot data was. Acquire both devices and let a Fusion-aware tool rebuild the logical volume.

Mac Acquisition

Fusion Drive

While the HDD can be easily visible after removing the glass, the SSD is hidden behind the motherboard and not always visible.



Mac acquisition decision tree

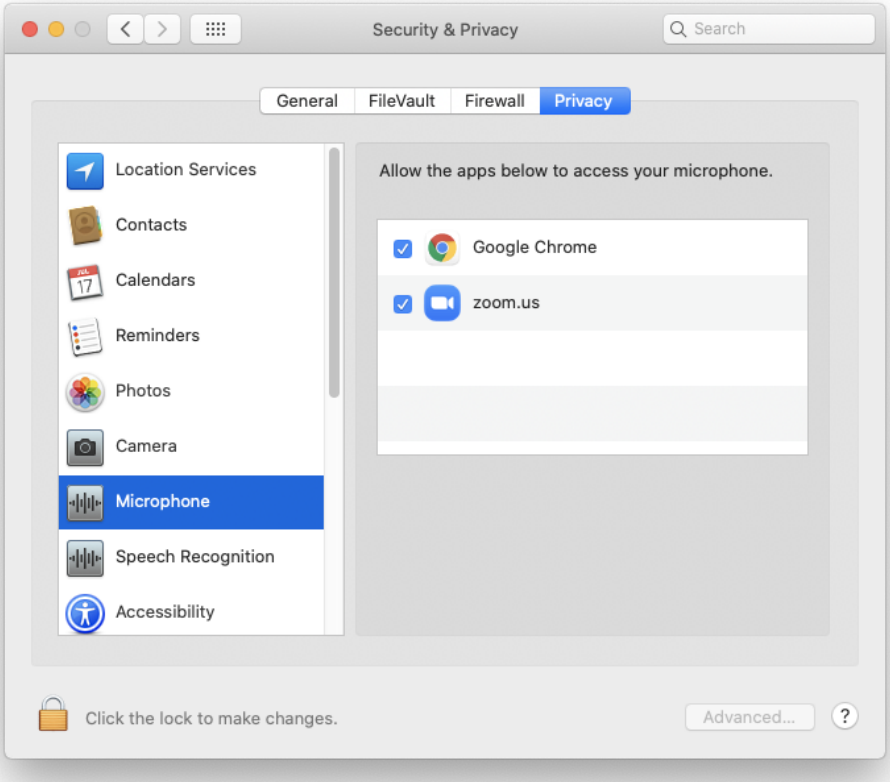
- **1. Running and unlocked?** Then the clock is ticking: capture RAM, running state and user-context artifacts before it sleeps or locks.
- **2. FileVault on?** Obtain the password, recovery key or escrowed key if legally available. Without one, a powered-off Mac is a brick.
- **3. Which generation?** Intel (pull the drive), T2 (boot it or live), Apple silicon (boot it or live, owner password needed for security changes).
- **4. What is the goal?** Full image, targeted collection, triage package or live-response snapshot. Scope decides tooling and time.
- **5. Can the tool handle it?** APFS containers, volume groups, snapshots, FileVault and Fusion. If not, you will miss data and not know it.

Live acquisition: what to grab, in order

- **Identity:** logged-in user, hostname, serial, model, chip, OS version and build, local time zone and current time (compare to a trusted clock).
- **Volatile state:** RAM image first, then running processes, network connections, mounted volumes, logged-in sessions, open files.
- **Security state:** FileVault status, Secure Token holders, SIP status, recovery key escrow, MDM enrolment.
- **User activity:** recent downloads, browser history, shell history, persistence locations, Biome and KnowledgeC, app databases.
- **Integrity:** hashes and tool logs for every package. Note every command you ran on the evidence Mac.

FileVault and hardware encryption

- **What FileVault proves:** without credentials or a recovery key the Data volume stays sealed. The System volume (read-only, Apple-signed) is never encrypted and holds nothing user-specific.
- **On T2 and Apple silicon** internal storage is always encrypted at the hardware layer; FileVault decides only whether the user password is required to unwrap the key at boot.
- **Decision rule:** a powered-on unlocked Mac beats a powered-off encrypted Mac, every time. Do not shut down to “preserve” it.
- **Document the output:** encrypted or decrypted, logical or physical, sparseimage, DMG, E01, AFF4 or targeted folder. Each limits what a later examiner can do.


Security & Privacy
Search
General
FileVault
Firewall
Privacy
Location Services
Contacts
Calendars
Reminders
Photos
Camera
Microphone
Speech Recognition
Accessibility
Allow the apps below to access your microphone.
Google Chrome
zoom.us
Click the lock to make changes.
Advanced...
?

Core macOS artifact map

Question	Artifact families to check
What did the user do?	KnowledgeC, Biome, ScreenTime, ARD, Recent Items, Saved State, Unified Logs.
What files were touched?	FSEvents, Spotlight, QuickLook, APFS metadata, Document Revisions, app databases.
What was downloaded?	Browser DBs, quarantine xattrs, LSQuarantine, Mail attachments, iCloud metadata.
What ran or persisted?	LaunchAgents, LaunchDaemons, Login Items, shell history, Gatekeeper, XProtect, Unified Logs.
What devices/networks were used?	Wi-Fi plists, Bluetooth, NetworkInterfaces.plist, mounted volume history, USB logs.

Who used this Mac? Users, accounts and identity

Where to look

- /private/var/db/dslocal/nodes/Default/users/*.plist: UID, GID, UUID, home path, shell, password hint, creation date.
- /Users/<name>: ownership and timestamps tie artifacts to a user context.
- Unified Logs (loginwindow, opendirectoryd) and KnowledgeC/Biome device-unlock streams for session timing.

What it can prove

- Which local accounts existed, when they were created, which groups (admin, Secure Token) they belonged to, who was logged in at a given time.

What it does not prove

- That a specific person was at the keyboard. Shared passwords, auto-login and screen-sharing sessions all break the account-to-person link.

Tip

- Correlate with Apple ID sign-in plists (MobileMeAccounts.plist) and iCloud device names.

What did they type? Shell and terminal artifacts

Where to look

- ~/.zsh_history (default shell since Catalina) and ~/.bash_history: one line per command, zsh adds an epoch timestamp when EXTENDED_HISTORY is on.
- ~/Library/Saved Application State/com.apple.Terminal.savedState/: window contents as of the last quit.
- ~/.ssh/known_hosts, config, keys; ~/.zsh_sessions/ for per-tab history on recent macOS.

What it can prove

- Commands the user ran, hosts they connected to, tools they installed, files they touched from the shell.

What it does not prove

- Order or time of commands when history has no timestamps; that the command succeeded; who typed it. History is user-editable and can be disabled (HISTSIZE=0, unset HISTFILE).

Tip

- iTerm2, Warp and VS Code keep their own terminal history.

What happened on disk? FSEvents

Where to look

– /.fseventsd/ at the root of every APFS or HFS+ volume, including external drives and Time Machine backups.

– gzip-compressed log files named by event ID; parse with FSEParser, mac_apt or your suite.

What it can prove

– That a path was created, modified, renamed, removed or had its permissions changed, with the event sequence, at directory granularity on recent macOS.

What it does not prove

– Exact timestamps (records are batched), file content, that a file was opened or read, or which user caused the change. The log also rolls over quickly on busy volumes.

Tip

– Best used with Spotlight, APFS inode times, app databases and Unified Logs to turn “something happened here” into “this happened then”.

What did the system see? Unified Logs

Where to look

- Replaced syslog/ASL in macOS 10.12 Sierra. Stored as .tracev3 under /private/var/db/diagnostics with UUID text under /private/var/db/uuidtext.
- Collect: **log collect --last 7d --output case.logarchive** Review: **log show --predicate 'process == "loginwindow"' case.logarchive**

What it can prove

- Process starts, authentication, screen lock/unlock, mounts and USB device arrivals, network changes, Gatekeeper and TCC decisions, Time Machine and iCloud sync, with millisecond timestamps.

What it does not prove

- Everything: retention is days to weeks, and sensitive fields are replaced by <private> unless a profile disabled redaction. Absence of a log line is not absence of the event.

Tip

- Example (simplified): 2026-10-13 21:14:31 loginwindow: user alex unlocked the screen. Matches the KnowledgeC and Biome example later.

What was installed? System and install logs

Where to look

- /var/log/install.log: OS updates, pkg installs, App Store installs, with installer name and user.
- /var/db/receipts/*.bom and .plist: what each package put on disk.
- /var/log/system.log and ASL-era logs on older Macs; audit logs only if an audit policy was configured.

What it can prove

- When software arrived, by which installer, what files it laid down, and when OS upgrades happened (useful for bounding artifact formats).

What it does not prove

- That the software ran, or who launched the installer, unless the log line names the user. Drag-and-drop .app installs leave no install.log entry at all.

Tip

- Correlate with quarantine xattrs, LaunchAgents and app usage in KnowledgeC/Biome.

What ran or kept running? Execution and persistence

Where to look

– LaunchAgents (~/Library and /Library), LaunchDaemons (/Library, /System/Library), login and background items (.btm), cron, periodic, shell startup files, Login Items in MDM profiles.

– Execution: KnowledgeC /app/inFocus and Biome App.InFocus, Unified Logs, Spotlight kMDItemLastUsedDate, Saved Application State, code-sign and XProtect logs.

What it can prove

– That a program was configured to start, when the persistence file was written, and that an app was in the foreground at a given time.

What it does not prove

– That a daemon actually ran at a given moment (check the logs), or that a user knowingly installed it. Many legitimate apps install agents.

Tip

– Ventura and later surface background items in System Settings > General > Login Items; the .btm file behind it is the artifact.

KnowledgeC: locations and value

- System context: /private/var/db/CoreDuet/Knowledge/knowledgeC.db
- User context: ~/Library/Application Support/Knowledge/knowledgeC.db
- It is the database behind Screen Time, Siri suggestions and “Continue on Mac”: a diary of what the device was doing, kept for about four weeks.
- Common streams: /app/inFocus (which app was front-most), /app/activity, /safari/history, /display/isBacklit (screen on), /device/isLocked, /media/nowPlaying, /app/install.
- **Status in 2026:** the file still exists on macOS 27, but since Ventura (2022) most streams are written to **Biome** first. Treat KnowledgeC as the legacy view and Biome as the primary one on modern systems.

KnowledgeC database structure

Table / field	What to review
ZOBJECT	Primary event table. Review ZSTREAMNAME, ZSTARTDATE, ZENDDATE, ZVALUESTRING, ZVALUEINTEGER and source fields.
ZSOURCE	Source app or service metadata, often helpful for bundle ID attribution.
ZSTRUCTUREDMETADATA	Extra event metadata, commonly a blob or archived object. Needs parser support.
ZSTREAMNAME	Event type such as /app/inFocus, /app/activity, /safari/history or /display/isBacklit.
Timestamps	Usually Mac Absolute time. Add 978307200 seconds to convert to Unix epoch.

KnowledgeC example: app in focus

	ENTRY CREATION	DAY OF WEEK	GMT OFFSET	START	END	USAGE IN SECONDS	ZSTREAMNAME	ZVALUESTRING
243	2018-05-13 18:06:33	Sunday	-7	2018-05-13 18:05:25	2018-05-13 18:06:33	68	/app/inFocus	ch.protonmail.protonmail
244	2018-05-13 20:55:47	Sunday	-7	2018-05-13 20:41:46	2018-05-13 20:55:47	841	/app/inFocus	com.apple.DocumentsApp
245	2018-05-13 20:55:48	Sunday	-7	2018-05-13 20:53:09	2018-05-13 20:55:48	159	/app/inFocus	com.apple.quicklook.extension.previewUI
246	2018-05-13 20:56:01	Sunday	-7	2018-05-13 20:55:55	2018-05-13 20:56:01	6	/app/inFocus	com.apple.MobileSMS
247	2018-05-13 20:58:20	Sunday	-7	2018-05-13 20:58:12	2018-05-13 20:58:19	7	/app/inFocus	com.apple.mobilemail
248	2018-05-13 20:58:50	Sunday	-7	2018-05-13 20:58:21	2018-05-13 20:58:50	29	/app/inFocus	com.starbucks.mystarbucks
249	2018-05-13 20:59:03	Sunday	-7	2018-05-13 20:58:52	2018-05-13 20:59:03	11	/app/inFocus	com.apple.Passbook
250	2018-05-14 09:23:22	Monday	-7	2018-05-14 09:23:16	2018-05-14 09:23:22	6	/app/inFocus	uk.co.bbc.news.notificationextension

KnowledgeC worked example: one

ZSTREAMNAME	ZSTARTDATE (raw)	Converted (PDT)	ZVALUESTRING	Plain English
/display/isBacklit	813644048	13 Oct 2026 21:14:08	1	Screen turned on
/device/isLocked	813644071	21:14:31	0	Someone unlocked the Mac
/app/inFocus	813644102	21:15:02	com.apple.Safari	Safari came to the front
/safari/history	813644200	21:16:40	https://example.com/invoice	Page visited in Safari
/app/inFocus	813645072	21:31:12	com.apple.Terminal	Terminal came to the front
/display/isBacklit	813645655	21:40:55	0	Screen went dark

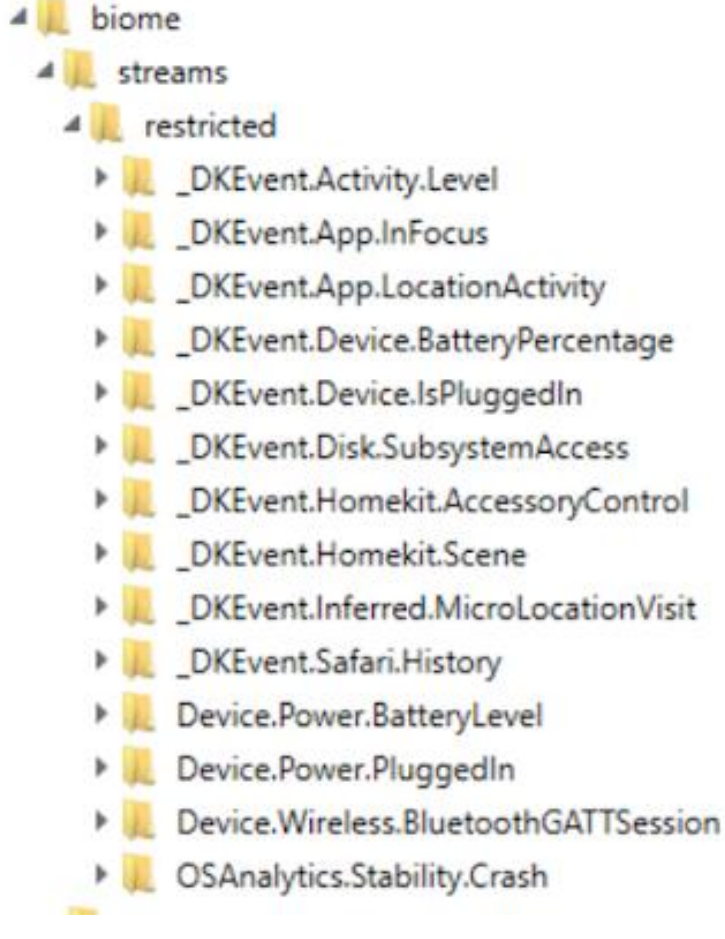
Convert: raw + 978,307,200 = Unix epoch. 813644048 + 978307200 = 1791951248 = 2026-10-14 04:14:08 UTC = 21:14:08 PDT (ZSECONDSFROMGMT = -25200). ZENDDATE minus ZSTARTDATE gives duration: Safari had focus for about 16 minutes.

KnowledgeC limitations and version differences

- Schemas and stream availability vary between macOS versions and between the system and user context databases.
- Some records are synced from other devices on the same Apple ID (look at ZSOURCE and device IDs), so a row is not automatically local activity.
- Retention is about four weeks. An empty database with a current timestamp usually means retention, not wiping.
- Since Ventura most streams are written to Biome first; by macOS 27 several stream names appear only in Biome. Always parse both and compare.

Biome data overview

- Biome stores activity streams in a folder hierarchy, commonly under /private/var/db/biome.
- Streams may be public or restricted and are often stored as SEGB binary files with metadata plists.
- High-value streams can include application focus, Safari history, device status, activity level and power state.
- Parsing is less mature than SQLite artifacts. Keep raw files for reprocessing as tools improve.



Biome interpretation notes

- Do not treat Biome as a single database. It is a stream-based artifact family.
- Check metadata.plist for stream properties such as retention/max age when present.
- SEGB filenames can contain Apple epoch-style timestamp values with sub-second precision.
- Expect rapid change across OS releases. Tool output should be validated against stream path and raw record timing.

Biome worked example

Stream folder	SEGB file (name = Cocoa time)	Record timestamp	Decoded payload	Plain English
_DKEvent.Device.IsLocked	813600000	813644071	isLocked = 0	Unlocked at 21:14:31
_DKEvent.App.InFocus	813600000	813644102 to 813645072	bundleID = com.apple.Safari	Safari front-most for 16 min 10 s
_DKEvent.Safari.History	813600000	813644200	URL + title	Invoice page visited 21:16:40
_DKEvent.App.InFocus	813600000	813645072 to 813645655	bundleID = com.apple.Terminal	Terminal front-most for 9 min 43 s
Device.Power.BatteryLevel	813600000	813645655	level = 0.41, charging = 0	Laptop, on battery, 41 %

Where: /private/var/db/biome/streams/ (system) and ~/Library/Biome/streams/ (user). Each stream folder holds SEGB files; the file name is the Cocoa timestamp of its first record (813600000 = 13 Oct 2026 09:00 PDT). Records inside are protobuf: use Biome parsers in iLEAPP/mac_apt or your suite.

QuickLook thumbnails

- QuickLook can create previews of files viewed in Finder or by applications.
- Common sources include thumbnail indexes and thumbnail data under user cache locations.
- Useful when a file is deleted, stored externally, encrypted elsewhere, or only briefly present.
- Do not overstate: a thumbnail suggests preview generation, not necessarily deliberate opening by a specific user.

Spotlight metadata

- .Spotlight-V100 stores metadata indexes for indexed volumes.
- Fields can include file names, paths, content hints, kMDItemWhereFroms, last used dates and document metadata.
- Great for finding files that existed, moved or were indexed on external volumes.
- Index state depends on volume settings, exclusions and whether the artifact was collected intact.

Recent items, Finder and Dock

- Recent items: .sfl2/.sfl3 lists, older recentitems plists, application-specific MRU lists.
- Finder preferences: window state, sidebar items, recent folders and sometimes view context.
- Dock plist: persistent apps, documents and recently used applications depending on configuration.
- Use these artifacts as navigation evidence: what was available or recently referenced, not proof of content.

Browser artifacts

- Safari: ~/Library/Safari/History.db, Downloads metadata, bookmarks, tabs/session state and website data.
- Chrome/Edge/Brave: History, Cookies, Login Data, Favicons, Top Sites, Preferences and extension data.
- Firefox: places.sqlite, cookies.sqlite, favicons.sqlite, sessionstore and downloads.
- Correlate history, downloads, quarantine, cached content and file system timestamps.

Safari History.db structure

Table	Common fields	Use
history_items	id, url, domain_expansion, visit_count	The URL identity and aggregate visit information.
history_visits	history_item, visit_time, title, load_successful, redirect_source	Timeline of visits and navigation chains.
metadata / segments	Version-dependent fields	Extra page metadata where present.
WAL	Uncheckpointed rows	May contain the most recent visits or deleted fragments.

Network, Wi-Fi and Bluetooth

- **Ventura and later:** /Library/Preferences/com.apple.wifi.known-networks.plist: SSID, BSSID history, first joined, last joined, added-by (user, MDM or iCloud sync).
- **Monterey and earlier:** /Library/Preferences/SystemConfiguration/com.apple.airport.preferences.plist.
- /Library/Preferences/SystemConfiguration/NetworkInterfaces.plist: interface inventory with MAC addresses.
- /Library/Preferences/com.apple.Bluetooth.plist: paired devices with names, addresses and last-connected times.
- Unified Logs give the best timing for joins, DHCP leases, VPN, captive portals and AirDrop sessions.

Keychain and credentials

- Keychains store passwords, certificates, Wi-Fi secrets, tokens and secure notes.
- Typical user source: ~/Library/Keychains; system source: /Library/Keychains/System.keychain.
- Decryption usually requires user password, key material or tool-specific support.
- Report credential artifacts carefully. Access to a secret does not automatically prove use.

Deleted data, APFS and SSD limits

- APFS, SSD TRIM and encryption reduce traditional unallocated-space recovery expectations.
- Snapshots, caches, QuickLook, Spotlight, app databases, cloud sync and backups often matter more than raw carving.
- Deleted SQLite rows may persist in WAL, freelists or journal fragments depending on timing and vacuuming.
- For reporting, explain whether recovery is from live file, snapshot, cache, database residue or carved fragment.

Cellebrite Digital Collector

- Born as **BlackBag MacQuisition** (2008), the first tool built to image Macs from a bootable Mac environment. Cellebrite acquired BlackBag in January 2020 and renamed it **Digital Collector** in 2021.
- One licensed USB device that does three jobs: **live acquisition** of a running Mac or Windows PC, **targeted collection** of selected files and artifacts, and **full forensic imaging** of the disk.
- Boots Intel, T2 and Apple silicon Macs from the Collector device, so the evidence Mac becomes the imaging station. Also runs as an app on a live system.
- Understands what generic imagers do not: APFS containers and snapshots, FileVault 2 (with password or recovery key), Fusion Drives, Core Storage and HFS+.
- Companion tool: **Cellebrite Inspector** (formerly BlackLight) for analysis of the collected image.

Digital Collector: what it can do

Capability	Mac	Windows	Note
RAM capture	Live	Live	First job on an unlocked machine
Targeted (logical) collection	Yes	Yes	Files, folders, artifact packs
Full disk image	Booted or live	Booted	E01, DMG or raw, hashed on the fly
Boot evidence Mac from Collector	Intel, T2, M1 to M3+	UEFI PCs	Apple silicon: reduced security plus owner password
FileVault / BitLocker	Password or recovery key	Recovery key	No key: you still get the encrypted image
APFS snapshots, Fusion, Core Storage	Yes	n/a	Rebuilds the Fusion logical volume
Latest documented release	3.8 (Oct 2024): Sequoia, M3. Autumn 2025: 4 TB devices		Verify macOS 26/27 and M4/M5 support in your release notes


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